Finite element analysis with AI

AgentFEM introductory exercises

Name _____ Date _____

Learning objectives

Describe a finite element problem, check modelling conditions and assess results using theory and numerical comparisons. No programming is required; identify the geometry, material, boundary conditions and quantities of interest.

Getting ready

Use an AI assistant connected to a local AgentFEM installation. First send:

Check whether you can run AgentFEM. Tell me its version and where projects will be saved. Ask for my approval before installing or configuring anything. For now, check the environment only; do not start a simulation.

Once available, send these instructions. Wait for the reply before sending an exercise.

I will send an exercise next. Once you receive it, create a new AgentFEM project, check and run it, and save the model and results. Show only one result figure, key values with units, and a two- or three-sentence explanation. Avoid lengthy analysis. Tell me if essential information is missing or you cannot run it. For now, reply only: ‘Ready. Please send the exercise.’ Do not start calculating or guess the exercise.

Procedure

1. Read the problem and predict the direction and magnitude of the response. Send the prompt, then check the figure, values and units.

2. Complete the comparison, fill in your observations and explain the results in your own words.

3. Check the reference answers on pages 8–10. If results differ substantially, check conditions before refining the mesh or time step.

Exercise guide

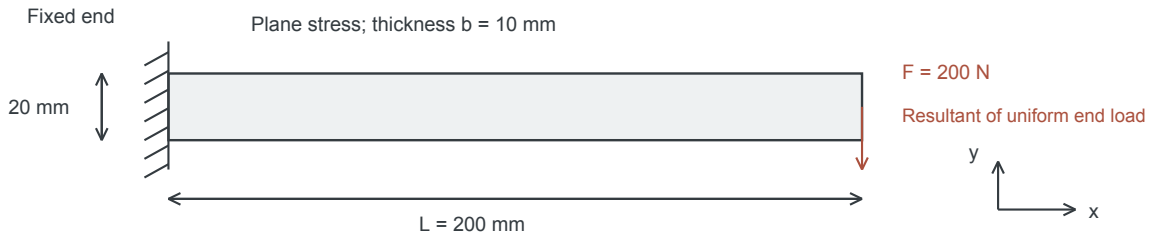
Page	Exercise	Check
2	Cantilever deflection	Beam theory and load scaling
3	Thick cylinder under internal pressure	Lamé solution
4	Steady heat conduction in a plate	Linear temperature and Sixier’s law
5	Transient heating of a plate	Analytical series and time-step comparison
6	Linear column buckling	Euler critical load
7	Re-entrant honeycomb	Solid plate comparison

Keep the model, a table of key values and a result figure for each exercise. Calculate in SI units and convert displayed values to mm, μm or °C as requested.

Setup: see the installation guide in the references. Windows uses WSL2.

Exercise 1 Cantilever deflection

Distinguish true displacement from display magnification and check load–displacement proportionality.



Problem conditions

Geometry	L = 200 mm; h = 20 mm; thickness b = 10 mm
Material	E = 210 GPa; ν = 0.30; homogeneous isotropic linear elasticity
Load and restraint	Left end fixed; total downward force 200 N uniformly distributed on the right end face
Assumptions	2D plane stress; small displacements; no gravity

Modelling prompt

Use AgentFEM to analyse a steel cantilever beam: length 200 mm, height 20 mm, thickness 10 mm, Young's modulus 210 GPa and Poisson's ratio 0.3. Fix the left end and distribute a total downward force of 200 N uniformly over the right end face. Use a 2D plane-stress, small-displacement linear-elastic model and neglect gravity. Show the deformed shape and the downward displacement at the midpoint of the right end (mm). State the deformation scale factor on the figure.

Tasks

1. Record the downward displacement at the right-end midpoint and check its direction.

2. Repeat at 400 N with all other conditions unchanged. Calculate the deflection ratio.

3. Compare with cantilever beam theory and discuss possible sources of discrepancy.

Comparison prompt

Increase the total force to 400 N, keep everything else unchanged and run again. Compare the two tip deflections and their ratio. Explain the result using beam theory.

Submit the deformed shape, a load–deflection table and a short comparison with theory.

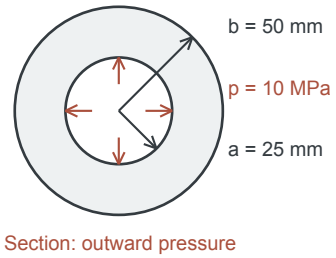
Observations

200 N: _____ mm 400 N: _____ mm Ratio: _____

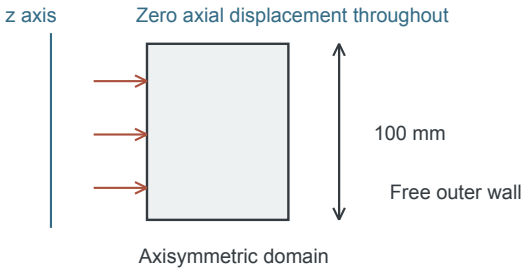
Theoretical deflection: _____ mm Relative difference: _____ %

Exercise 2 Thick cylinder under internal pressure

Use an axisymmetric model and check radial displacement against Lamé's solution.



Section: outward pressure



Problem conditions

Geometry	Inner radius a = 25 mm; outer radius b = 50 mm; length 100 mm
Material	E = 210 GPa; ν = 0.30; homogeneous isotropic linear elasticity
Load	Internal pressure p = 10 MPa; zero external pressure; no end-cap load
Assumptions	Axisymmetry; zero axial strain; axial displacement u_z = 0 throughout

Modelling prompt

Use AgentFEM to analyse a thick-walled cylinder: inner radius 25 mm, outer radius 50 mm, length 100 mm, Young's modulus 210 GPa and Poisson's ratio 0.3. Apply 10 MPa internal pressure, leave the outer wall radially free and apply no end-cap load. Use a linear-elastic axisymmetric model with zero axial displacement throughout. Report the inner-wall radial displacement (μm) and plot radial displacement through the wall at mid-length. Check against Lamé's solution.

Tasks

1. Check the directions of internal pressure and radial displacement.

2. Increase the outer radius to 60 mm. Keep inner radius and pressure fixed; compare inner-wall displacement.

3. Calculate analytical displacements for both radii and explain the effect of wall thickness.

Comparison prompt

Increase the outer radius to 60 mm, keep everything else unchanged and run again. Compare the two inner-wall radial displacements, check them against Lamé's solution and explain the effect of a thicker wall.

Submit the radial displacement profile with the analytical comparison and a table for both wall thicknesses.

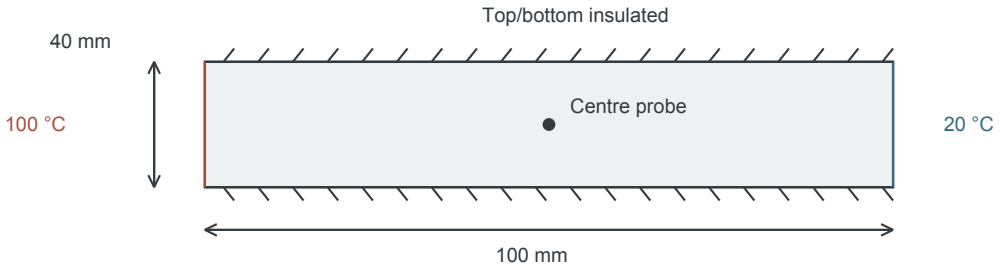
Observations

Outer radius 50 mm: _____ μm 60 mm: _____ μm

Reduction in inner-wall displacement: _____ %

Exercise 3 Steady heat conduction in a plate

Distinguish prescribed-temperature and insulated boundaries, and temperature from heat flux.



Problem conditions

Geometry and material	100 mm × 40 mm; uniform k = 45 W/(m·K)
Temperatures	Left 100 °C; right 20 °C
Other conditions	Other boundaries insulated; no internal heat source; 2D steady conduction

Modelling prompt

Use AgentFEM for 2D steady-state heat conduction in a metal plate: length 100 mm, height 40 mm and uniform thermal conductivity 45 W/(m·K). Hold the left boundary at 100 °C and the right at 20 °C. Insulate all other boundaries and use no internal heat source. Show the temperature field, centre temperature (°C), temperature profile along the horizontal centreline and rightward heat flux density (W/m²).

Tasks

1. Record the centre temperature and describe the profile along the length.

2. Double conductivity to 90 W/(m·K), without changing the boundaries. Compare temperatures and heat flux.

3. Derive the temperature profile from the 1D steady heat equation and use Sixier’s law to calculate heat flux.

Comparison prompt

Increase thermal conductivity to 90 W/(m·K), keep everything else unchanged and run again. Compare the temperature fields and heat flux densities. Explain the result using Fourier’s law.

Submit the temperature field, a table of centre temperature and heat flux for both conductivities, and an analytical check.

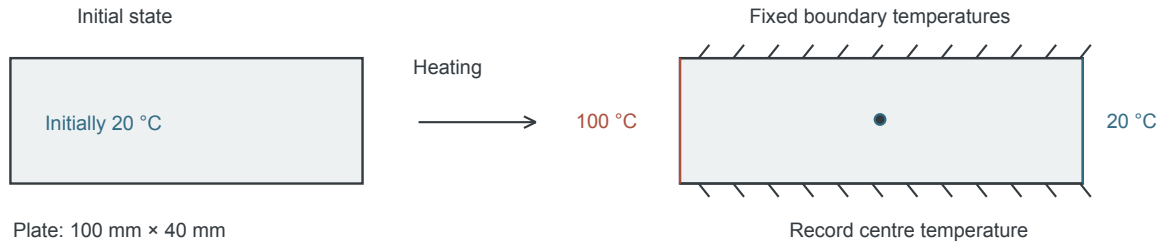
Observations

k = 45: centre _____ °C; heat flux _____ W/m²

k = 90: centre _____ °C; heat flux _____ W/m²

Exercise 4 Transient heating of a plate

Read a temperature history, compare transient and steady states, and test time-step sensitivity.



Problem conditions

Geometry	100 mm × 40 mm plate
Material	k = 45 W/(m·K); ρ = 7800 kg/m³; c = 500 J/(kg·K), all constant
Initial and boundary conditions	Initially 20 °C throughout; for t > 0, left 100 °C and right 20 °C
Other conditions	Other boundaries insulated; no internal heat source
Time integration	To 600 s; backward Euler; initial time step Δt = 2 s

Modelling prompt

Use AgentFEM to simulate transient heating of a 2D metal plate: length 100 mm, height 40 mm and initial temperature 20 °C. Keep thermal conductivity 45 W/(m·K), density 7800 kg/m³ and specific heat capacity 500 J/(kg·K) constant. From the start of heating, hold the left boundary at 100 °C and the right at 20 °C. Insulate all other boundaries and use no internal heat source. Run to 600 s using backward Euler with a 2 s time step. Provide the centre-temperature curve and data table (°C). Save the temperature field every 10 s as a time series playable in ParaView.

Tasks

1. Plot centre temperature and record values at 60, 120, 300 and 600 s.

2. Repeat with a 1 s time step and the same mesh. Compare temperatures at shared sample times.

3. Compare with Exercise 3 and justify whether the plate is sufficiently close to steady state at 600 s.

Comparison prompt

Reduce the time step to 1 s, keep the mesh and all other conditions unchanged and run again. Overlay the two centre-temperature curves. Compare the maximum temperature difference at shared sample times and the difference at 600 s to assess whether the original time step is adequate.

Submit both temperature histories, the four-time table, time-series files and your time-step assessment.

Observations

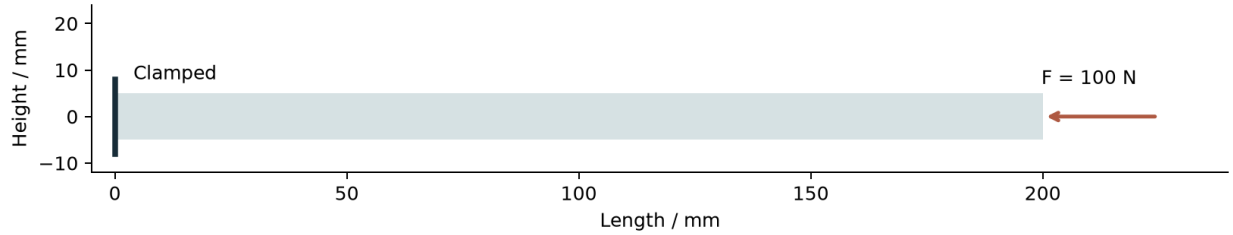
60 s: _____ °C 120 s: _____ °C

300 s: _____ °C 600 s: _____ °C

Maximum difference between time steps: _____ °C

Exercise 5 Linear column buckling

A lateral force bends a bar. Axial compression can make it buckle. Find the load at which sideways buckling begins.



Requires AgentFEM with the 2026-10-10 buckling update and SLEPc; v0.4.1 predates it.

Modeling prompt

Use AgentFEM for linear buckling of a steel column: length 200 mm, section height 10 mm and thickness 10 mm, Young’s modulus 210 GPa, Poisson’s ratio 0.3. Use 2D plane stress. Clamp the left end and apply uniform axial compression to the right face with a reference total force of 100 N; leave other boundaries free and neglect gravity. Solve the prestress state, then report the first critical load (N) and mode plot. Check against Euler’s cantilever-column formula and state that mode amplitude is not physical displacement. Set the eigenvalue solver tolerance to 1e-7 and check the residual.

Follow-up calculation

Increase the column length to 400 mm, keeping the section, material and supports unchanged. Recompute the first critical load, compare the two loads and explain their ratio using Euler’s formula.

Observations

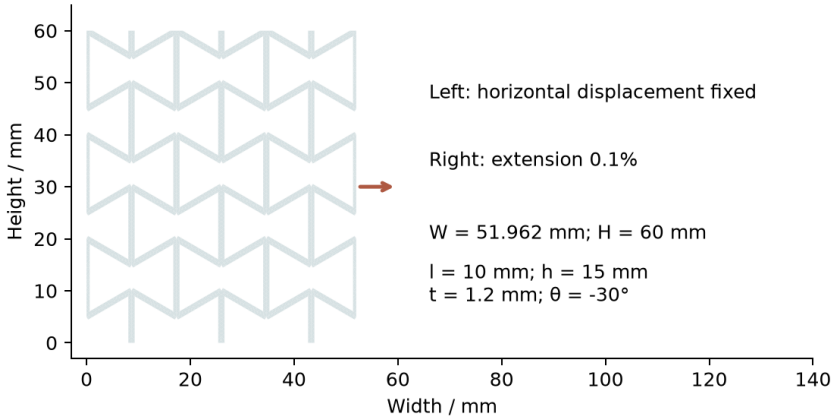
200 mm: _____ N 400 mm: _____ N Ratio: _____

Euler reference: _____ N Relative difference: _____ %

Submit one comparison plot, key values and a two- or three-sentence explanation.

Exercise 6 Tension of a re-entrant honeycomb

A solid plate usually narrows when stretched. A re-entrant honeycomb made of the same material can widen. Download and unzip the mesh pack, then give the folder to your AI assistant.



Download and unzip the pack; give the folder containing honeycomb.msh to the AI. Coordinates are in metres.

https://lab.haoming-luo.com/learn/files/honeycomb.zip

Modeling prompt

Use AgentFEM to import the attached honeycomb.msh without redrawing the geometry. Coordinates are in metres; width 51.9615 mm, height 60 mm. Boundary tags 11/12/13/14 mean left/right/bottom/top. Use small-strain linear elasticity in 2D plane stress, E=1 GPa and Poisson's ratio 0.3; neglect gravity. Fix horizontal displacement on the left and impose a horizontal extension of 0.1% of the width on the right. Leave vertical motion free except for the bottom-left vertical displacement. Use mean displacements on all four boundaries to calculate axial and transverse strains and apparent Poisson's ratio. Compare with a solid plate of the same size, material and constraints. Show one deformation comparison plot with its magnification.

Follow-up calculation

Halve the imposed horizontal extension for both specimens, leaving all other settings unchanged. Compare the apparent Poisson's ratios and explain why displacement decreases but the ratio barely changes.

Observations

Honeycomb: axial strain _____ transverse strain _____ ratio _____

Solid plate ratio: _____ Honeycomb ratio at half extension: _____

Submit one comparison plot, key values and a two- or three-sentence explanation.

Reference answers 1

Complete your observations before checking. Use these values to verify magnitudes and trends; different meshes need not agree digit for digit.

1 Cantilever deflection

For a rectangular section and an Euler–Bernoulli cantilever:

$I = bh^3/12 = 6.667 \times 10^{-9}\ m^4$; $\delta = FL^3/(3EI)$.

Total force	Beam theory / mm	FE or linear scaling / mm
200 N	0.381	0.383 (FE)
400 N	0.762	0.766 (linear scaling)

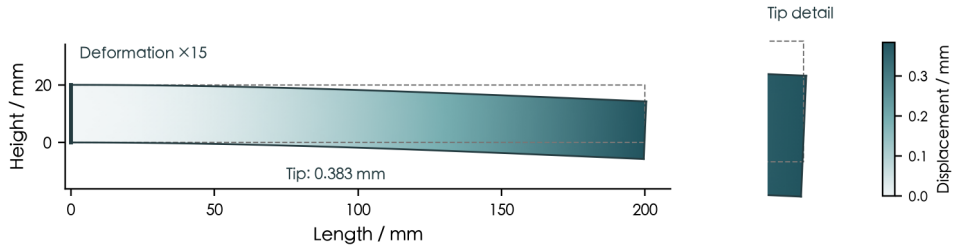


Figure 1. Beam displacement; dashed outline: original shape; deformation ×15.

Downward displacement is negative when y points upwards; the table reports its magnitude. The FE value is about 0.59% above beam theory. The 2D elasticity solution includes shear and end-restraint effects, whereas beam theory is approximate. Doubling the load gives a displacement ratio of 2 in linear elasticity.

2 Thick cylinder under internal pressure

For zero axial strain and zero external pressure, Lamé's solution gives:

$A = pa^2/(b^2 - a^2)$; $B = pa^2b^2/(b^2 - a^2)$.

$u_r(r) = [(1 + \nu)/E]\,[(1 - 2\nu)Ar + B/r]$.

Outer radius	Inner displacement / μm	Source
50 mm	2.270	Analytical and baseline FE
60 mm	2.003	Analytical

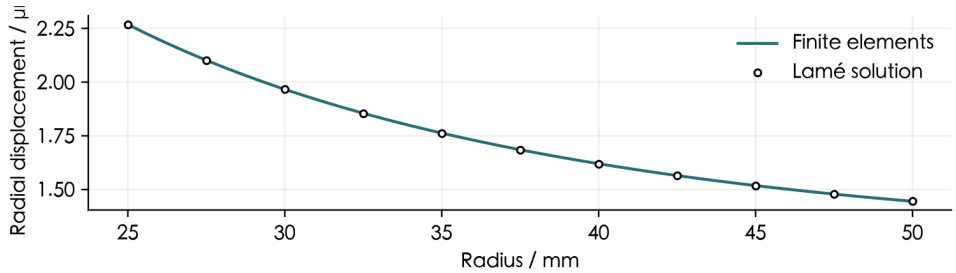


Figure 2. Radial displacement through the wall and Lamé's solution.

Increasing the outer radius from 50 to 60 mm reduces inner-wall displacement by about 11.76%. The axial restraint is unchanged. Do not apply this displacement formula directly to a freely extending or end-capped cylinder.

Reference answers 2

3 Steady heat conduction in a plate

With uniform material, no internal source and insulated top and bottom boundaries, temperature depends only on x. Set L = 0.1 m, with x increasing to the right:

$T(x) = 100 - 80x/L$ (°C); $q_x = -k\,dT/dx = 80k/L$.

Conductivity / W/(m·K)	Centre / °C	Heat flux / W/m²
45	60	36 000
90	60	72 000

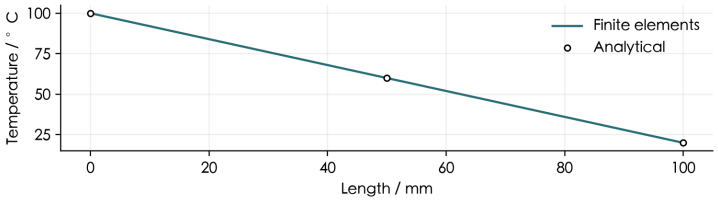


Figure 3. Temperature along the horizontal centreline.

At fixed end temperatures, changing uniform conductivity does not change this temperature profile, but it changes heat flux. Positive flux points to the right.

4 Transient heating of a plate

Thermal diffusivity $\alpha = k/(\rho c) = 1.154 \times 10^{-5}$ m²/s. The analytical centre temperature for $t > 0$ is:

$$T(L/2,t) = 60 - (160/\pi)\,\Sigma_{n=1}^{\infty}\,[\sin(n\pi/2)/n]\,\exp(-n^2\pi^2\alpha t/L^2).$$

Time / s	Δt = 2 s / °C	Δt = 1 s / °C	Analytical / °C
60	34.147	34.231	34.319
120	46.814	46.914	47.014
300	58.263	58.296	58.328
600	59.941	59.943	59.945

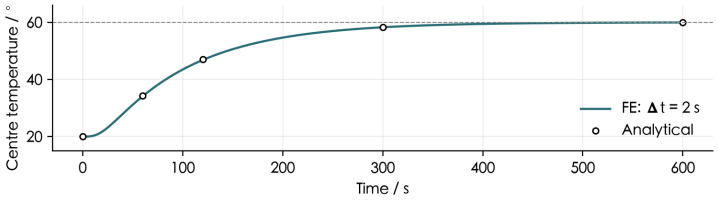


Figure 4. Centre temperature history; dashed line: steady state at 60 °C.

The maximum difference at shared sample times is 0.133 °C; at 600 s it is 0.0022 °C. Agreement at the endpoint alone is insufficient: check the whole history against the required accuracy.

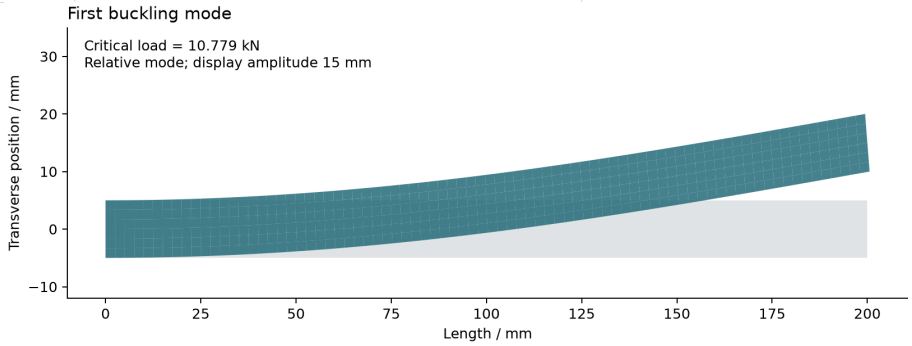
Reference runs: AgentFEM 0.4.0.dev0 / DOLFINx 0.11.0. Beam: 80 × 8 quadratic elements; cylinder: 40 × 4 quadratic elements; heat: 80 × 16 linear elements. Backward Euler for transient heat; values converted to the displayed units.

Reference answers 3

5 Column buckling

$I = bh^3/12$; $P_{cr} = \pi^2 EI/(4L^2)$.

Length / mm	FE / N	Euler / N
200	10779.48	10794.88
400	2698.18	2698.72

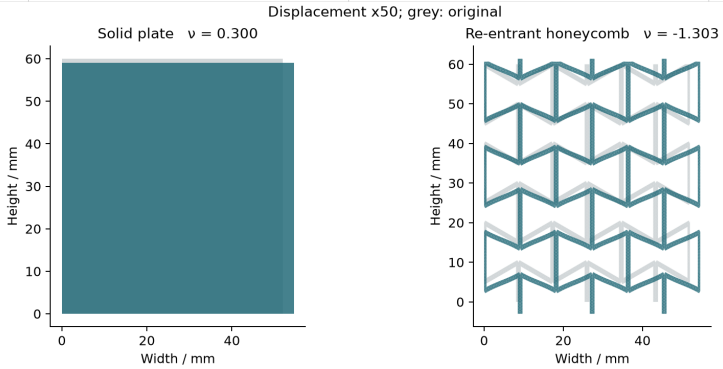


Mode amplitude is for display only. Critical load equals reference force times eigenvalue; the long-to-short column load ratio is about 0.25.

6 Negative Poisson's ratio

$\nu_{app} = -\varepsilon_y/\varepsilon_x$.

Specimen	Axial strain	Transverse strain	Apparent ratio
Honeycomb	0.001000	0.001303	-1.3026
Solid plate	0.001000	-0.000300	0.3000



Halving the extension leaves the apparent ratio unchanged. Refining the honeycomb mesh from 0.8 to 0.4 mm changes it from −1.3026 to −1.3062 (about 0.28%). The negative value comes from geometry, not the material parameter. A finite-specimen response is not a periodic effective property.

References

[1] Interactive web edition with copyable prompts and results. https://lab.haoming-luo.com/learn/en.html

[2] AgentFEM installation guide. https://github.com/haoming-luo/agentfem/blob/main/INSTALL.md

Acknowledgments: We thank Professor Meng Wang of Beijing Institute of Technology for first proposing the use of AI-native finite-element simulation in the classroom, and Renzi Bai of Northwestern Polytechnical University for completing the first validated AgentFEM installation on Windows and contributing to this handout.